



MAJOR PROJECT REPORT

on

CS Shield: Defensive Monitoring System for SCADA Networks

Using OpenPLC, Node-Red, Suricata, Zeek IDS & Wazuh SIEM

Submitted in partial fulfillment of the requirements for the

Degree of Bachelor of Technology

in

Computer Science & Engineering (CyberSecurity & CyberDefence)

Faculty of Engineering and Technology

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Declaration

We hereby declare that the Major Project titled "CS Shield: Defensive Monitoring System for SCADA Networks using OpenPLC, Suricata IDS & Wazuh SIEM" submitted to the Faculty of Engineering and Technology, Sri Sri University, is a record of original work done by us under the guidance of Ishita Kosambia.

The work embodied in this report has not been submitted for the award of any other degree or diploma at this or any other institution.

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Certificate

This is to certify that the Major Project titled "CS Shield: Defensive Monitoring System for SCADA Networks using OpenPLC, Suricata IDS & Wazuh SIEM" has been carried out by Nisha, Shivam, Aloukik, Ved, Arun, Roll No. FET-BCD-2023-27-011, FET-BCD-2023-27-012, FET-BCD-2023-27-014, FET-BCD-2023-27-029, FET-BCD-2023-27-031, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering from Sri Sri University.

The project work is original and satisfactory, and the students have shown adequate aptitude in the design, development, and analysis of the system.

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Abstract

Supervisory Control and Data Acquisition (SCADA) systems and Industrial Control Systems (ICS) are increasingly becoming targets of sophisticated cyberattacks. The convergence of Operational Technology (OT) with Information Technology (IT) networks has introduced significant security vulnerabilities to critical infrastructure, including power plants, water treatment facilities, oil and gas industries, and manufacturing plants.

This project presents the design and implementation of CS Shield — a lightweight SCADA security monitoring framework developed using open-source tools. The framework integrates OpenPLC as the industrial PLC simulator, Suricata as the network-level Intrusion Detection System (IDS), and Wazuh as the centralized Security Information and Event Management (SIEM) platform.

OpenPLC simulates industrial control operations such as water tank process control using Modbus TCP communication. Suricata monitors network traffic and performs signature-based intrusion detection. Wazuh collects logs, generates alerts, and provides centralized visualization through dashboards.

The combined solution enables detection of network anomalies, unauthorized access attempts, port scanning activities, and suspicious industrial protocol traffic through custom detection rules and automated alerting. The Wazuh dashboard provides a unified interface for monitoring, visualization, and incident response.

The results demonstrate that the integrated CS Shield framework successfully detects Nmap scans, unauthorized Modbus communication attempts, and other suspicious activities, while maintaining low CPU and memory overhead on the underlying systems.

Keywords:

SCADA Security, ICS, OpenPLC, Suricata, Wazuh, SIEM, IDS, Modbus TCP, Network Monitoring, Intrusion Detection, PLC Simulation, Industrial Cybersecurity

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Chapter 1: Introduction

1.1 Background of SCADA Security

Supervisory Control and Data Acquisition (SCADA) systems are industrial control systems used to monitor and control critical infrastructure such as power plants, water treatment facilities, oil and gas industries, manufacturing plants, and transportation systems. These systems consist of Programmable Logic Controllers (PLCs), sensors, actuators, Human Machine Interfaces (HMIs), and communication networks.

Traditionally, SCADA systems operated in isolated environments with limited external connectivity. However, with the rise of Industrial Internet of Things (IIoT), cloud integration, and remote access technologies, SCADA systems are increasingly connected to enterprise networks and the internet. This connectivity has improved operational efficiency but has also introduced significant cybersecurity risks.

Cyber attacks targeting industrial systems have increased in recent years. Incidents such as Stuxnet, Triton, BlackEnergy, and Industroyer demonstrated that attackers can disrupt industrial processes, manipulate equipment, and damage critical infrastructure. As a result, monitoring and securing industrial environments have become essential.

SCADA systems are usually designed with a focus on operational availability rather than cybersecurity. Many industrial networks lack proper intrusion detection, centralized monitoring, and threat visibility. Therefore, lightweight monitoring frameworks are required to improve security without affecting industrial operations.

1.2 Problem Statement

Most SCADA environments face multiple security challenges because they were not originally designed to resist cyber attacks. Industrial systems often use outdated software, insecure communication protocols, and limited authentication mechanisms.

Traditional monitoring systems have several limitations:

- High deployment cost
- Complex infrastructure requirements
- Limited support for industrial environments
- Lack of centralized monitoring
- Difficulty detecting suspicious traffic

Many small and medium-scale industrial organizations cannot afford expensive industrial security solutions. As a result, attacks such as unauthorized access, reconnaissance scanning, and malicious network activities may remain undetected.

There is a need for a lightweight, cost-effective, and open-source security monitoring framework capable of:

- Monitoring industrial network traffic
- Detecting suspicious activities

- Generating alerts
- Centralizing logs
- Providing dashboard visualization

1.3 Objective of the Project

The primary objective of the project is to develop a lightweight SCADA security monitoring framework named CS Shield using open-source tools.

The project objectives include:

- Monitoring industrial network traffic
- Detecting suspicious activities in SCADA environments
- Integrating intrusion detection systems
- Generating alerts for abnormal behavior
- Collecting and analyzing logs
- Visualizing events using dashboards
- Simulating attacks for security testing
- Monitoring PLC communication in real time

1.4 Overview of CS Shield

CS Shield is a lightweight SCADA monitoring framework developed using:

- OpenPLC
- Suricata
- Wazuh
- Nmap
- iperf3
- Ubuntu Virtual Machines

The framework simulates a small industrial control environment where OpenPLC acts as the PLC controller generating industrial traffic. Suricata monitors network traffic and performs intrusion detection using signature-based rules. Wazuh collects logs, generates alerts, and provides centralized visualization through dashboards.

The system performs: network traffic monitoring, PLC communication monitoring, intrusion detection, alert generation, event visualization, and attack simulation analysis.

1.5 Scope and Limitations

Scope

The scope of the project includes:

- SCADA network traffic monitoring
- Intrusion detection using Suricata
- Centralized log monitoring using Wazuh

- OpenPLC industrial process simulation
- Detection of suspicious activities
- Visualization of alerts and logs
- Simulated attack analysis

Limitations

The project has certain limitations:

- Implemented in a virtualized environment
- Uses mainly signature-based detection
- Limited industrial protocol analysis
- No machine learning integration
- Limited scalability testing
- Simulated industrial setup only

1.6 Role of OpenPLC in the Project

OpenPLC is an open-source Programmable Logic Controller platform used to simulate industrial control processes in the SCADA environment.

In the CS Shield framework, OpenPLC simulates industrial operations such as: water tank process control, sensor monitoring, pump control logic, and PLC register updates.

OpenPLC generates industrial traffic using Modbus TCP communication, allowing Suricata and Wazuh to monitor industrial activities and detect suspicious behavior. The integration of OpenPLC provides realistic industrial communication, PLC simulation capability, industrial traffic generation, process monitoring environment, and a SCADA attack testing platform.

Chapter 2: Critical Review

2.1 Existing SCADA Security Approaches

Modern SCADA security solutions include firewalls, Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Security Information and Event Management (SIEM), Threat Intelligence Platforms, network segmentation, and access control systems. Large industrial organizations use advanced monitoring systems to detect attacks and secure operational technology environments.

2.2 Limitations of Traditional Monitoring

Traditional security systems have several drawbacks: expensive deployment, high maintenance cost, complex configuration, enterprise-focused designs, limited industrial protocol support, and lack of lightweight deployment options. These limitations make deployment difficult for small industrial environments.

2.3 Need for Lightweight Monitoring

Industrial environments require lightweight monitoring solutions because SCADA devices have limited resources, operational availability must be maintained, industrial systems cannot tolerate heavy monitoring overhead, and smaller organizations require affordable solutions. Lightweight monitoring frameworks provide essential visibility without disrupting industrial operations.

2.4 Improvement through CS Shield

CS Shield improves industrial monitoring by using open-source tools, providing low-cost deployment, supporting centralized monitoring, detecting suspicious traffic, visualizing events through dashboards, and supporting industrial traffic analysis.

Chapter 3: Data Handling and Traffic Monitoring

3.1 Types of Data

The project handles two primary categories of data. Network Traffic includes: TCP/IP packets, Modbus TCP communication, network scans, and simulated attack traffic. Log Data is generated from: Suricata alerts, Wazuh logs, system monitoring events, and OpenPLC activities.

3.2 Traffic Monitoring using Suricata

Suricata is an open-source Intrusion Detection and Prevention System (IDPS) used for network traffic monitoring. Features used in the project include: packet inspection, signature matching, protocol analysis, alert generation, and real-time traffic monitoring. Suricata analyzes network packets and identifies suspicious traffic patterns.

3.3 Log Handling

Logs generated by Suricata and system components are forwarded to Wazuh for centralized monitoring. Log handling operations include: collection, parsing, storage, correlation, and visualization.

3.4 Basic Threat Intelligence (ThreatFox)

ThreatFox provides Indicators of Compromise (IOCs) such as malicious IP addresses, malware domains, and threat signatures. These indicators improve threat detection capabilities.

3.5 OpenPLC Traffic Generation

OpenPLC generates industrial communication traffic within the SCADA environment. The PLC continuously exchanges data related to water level values, pump control status, sensor readings, and PLC register updates. This traffic is analyzed by Suricata for abnormal activities.

3.6 Industrial Protocol Monitoring

The project monitors industrial communication protocols such as Modbus TCP and TCP/IP communication. Suricata captures and analyzes industrial traffic to identify unauthorized requests, suspicious scanning, and abnormal communication patterns.

Chapter 4: Detection Mechanism

4.1 Intrusion Detection using Suricata

Suricata performs intrusion detection by analyzing packet contents and comparing traffic against predefined rules. Detection includes: port scanning, unauthorized access attempts, suspicious TCP communication, and abnormal network behavior.

4.2 Signature-based Detection

Signature-based detection identifies threats by matching traffic patterns against known attack signatures. Advantages include fast detection, easy deployment, and low overhead. Disadvantages include inability to detect unknown attacks and requirement for regular rule updates.

4.3 Rule Configuration

Custom Suricata rules were configured to detect suspicious activities. Example rule:

```
alert tcp any any -> any any (msg:"Nmap Scan Detected"; sid:1000001;)
```

4.4 Log Monitoring using Wazuh

Wazuh monitors logs generated by Suricata and system components. Functions include: alert analysis, event correlation, dashboard visualization, and security monitoring.

4.5 Alert Generation

Alerts are generated whenever suspicious activities are detected. Alert information includes: timestamp, source IP, destination IP, severity level, and alert description.

4.6 Monitoring OpenPLC Communication

Suricata continuously monitors communication involving OpenPLC. Monitoring includes packet inspection, Modbus traffic analysis, TCP session monitoring, and alert generation.

4.7 Detection of Unauthorized PLC Access

The system detects suspicious attempts to interact with PLC services. Examples include unauthorized Modbus requests, port scans against PLC services, and excessive connection attempts. Example rule:

```
alert tcp any any -> any 502 (msg:"Possible Modbus TCP Activity Detected";  
sid:1000002;)
```

Chapter 5: System Setup

5.1 Ubuntu VM Environment

The project was implemented using Ubuntu virtual machines. Advantages include: open-source environment, easy deployment, lightweight operation, and flexible configuration.

5.2 Tools Used

Tool	Purpose
OpenPLC	PLC simulation, industrial process control, Modbus TCP communication
Suricata	Intrusion detection, traffic analysis, alert generation
Wazuh	Log management, alert visualization, centralized SIEM
Nmap	Attack simulation, network reconnaissance testing
iperf3	Traffic generation, performance testing
Ubuntu VMs	Virtualized deployment environment

5.3 Basic Network Setup

The environment consists of three virtual machines: the OpenPLC VM (industrial PLC simulator), the Monitoring VM (running Suricata and Wazuh agent), and the Attacker VM (used for simulated reconnaissance). Traffic flows through the monitored network where Suricata analyzes packets and forwards logs to Wazuh.

5.4 OpenPLC Setup

The OpenPLC runtime was installed on Ubuntu virtual machines. The setup included:

1. Installing OpenPLC Runtime
2. Configuring network interfaces
3. Uploading PLC logic
4. Enabling Modbus TCP service
5. Verifying communication

The PLC logic simulated a water tank process with water level monitoring, pump control, and sensor communication.

Chapter 6: Implementation

6.1 Installation & Configuration

Implementation steps included:

6. Ubuntu installation on all virtual machines
7. Suricata installation and rule configuration
8. Wazuh server and agent setup
9. OpenPLC configuration
10. Network interface setup
11. Log forwarding configuration

6.2 Monitoring Setup

Suricata was configured to capture packets, monitor interfaces, and generate alerts. Wazuh was configured for log collection, alert visualization, and event monitoring.

6.3 Attack Simulation using Nmap

Nmap scans were used to simulate reconnaissance attacks. Example commands:

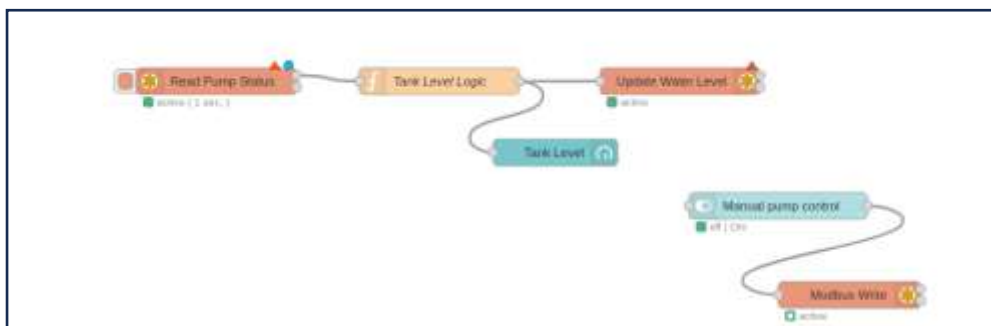
```
nmap -ss <target-ip>
nmap -sV <OpenPLC-IP>
```

6.4 Performance Testing using iperf3

iperf3 generated network traffic to test throughput, stability, and monitoring performance. The lightweight framework maintained stable performance during testing.

6.5 OpenPLC Implementation

OpenPLC simulated: water tank level monitoring, pump control, sensor communication, and PLC register updates. Industrial traffic generated by OpenPLC was monitored by Suricata and Wazuh.



6.6 PLC Register Monitoring

Registers monitored included the water level register, pump status register, and alarm register. The monitoring system analyzed communication involving these registers.

6.7 Attack Simulation on OpenPLC

Attack simulations targeted the OpenPLC web interface, Modbus TCP service, and open ports. Example command:

```
nmap -sS -p 502 <OpenPLC-IP>
```

Suricata successfully detected these scans and generated corresponding alerts.

```
test@OpenPLC-VH:~/zeek_logs$ sudo chown $USER modbus.log
test@OpenPLC-VH:~/zeek_logs$ cat modbus.log | grep 192.168.200.143
1775584069.769355 C2x6EAUW0jnTaakf2 192.168.200.137 55690 19
2.168.200.143 502 1 1 WRITE_SINGLE_COIL REQ -
1775584184.071434 Cu4u0Y14aE4qThMoIf 192.168.200.137 56186 19
2.168.200.143 502 1 1 WRITE_SINGLE_COIL REQ -
1775584242.938047 C2YFwj1eRPbQWmdhc 192.168.200.137 33060 19
2.168.200.143 502 1 1 WRITE_SINGLE_REGISTER REQ -
1775584284.197535 CMIOY22CTx6xlFL64 192.168.200.137 38500 19
2.168.200.143 502 1 1 WRITE_SINGLE_REGISTER REQ -
1775584361.859759 CoFMSK2PHubGwMFIRe 192.168.200.137 47450 19
2.168.200.143 502 1 1 WRITE_SINGLE_REGISTER REQ -
test@OpenPLC-VH:~/zeek_logs$
```

```
test@OpenPLC-VH:~/zeek_logs$ zeek-cut id.orig_h id.resp_h function_code va
lue < modbus.log
192.168.200.137 192.168.200.143
192.168.200.137 192.168.200.143
192.168.200.137 192.168.200.143
192.168.200.137 192.168.200.143
192.168.200.137 192.168.200.143
test@OpenPLC-VH:~/zeek_logs$
```

```
{
  "timestamp": "2020-04-07T22:20:49.060548+0530",
  "flow_id": 2059313200300999,
  "in_iface": "ens33",
  "event_type": "alert",
  "src_ip": "192.168.200.137",
  "src_port": 62579,
  "dest_ip": "192.168.200.143",
  "dest_port": 502,
  "proto": "TCP",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 100000,
    "rev": 1,
    "signature": "Malformed Modbus Packet",
    "category": "",
    "severity": 3
  },
  "flow": {
    "packets_received": 2,
    "packets_transmitted": 1,
    "bytes_received": 120,
    "bytes_transmitted": 50,
    "start": "2020-04-07T22:20:49.060503+0530"
  }
}
```

```
{
  "timestamp": "2020-04-07T23:40:54.249575+0530",
  "flow_id": 1516871999212679,
  "in_iface": "ens33",
  "event_type": "alert",
  "src_ip": "192.168.200.137",
  "src_port": 34862,
  "dest_ip": "192.168.200.143",
  "dest_port": 502,
  "proto": "TCP",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 100002,
    "rev": 1,
    "signature": "MODBUS Write Single Register (FC06)",
    "category": "",
    "severity": 3
  }
}
```

Chapter 7: Results

7.1 Alerts Generated

The framework generated alerts for: port scans, suspicious TCP traffic, unauthorized access attempts, and PLC-related suspicious activities.

7.2 Log Observation in Wazuh

Logs displayed: source IPs, destination IPs, alert categories, and event timestamps. The Wazuh dashboard provided centralized visibility into all security events.

7.3 Detection of Suspicious Activity

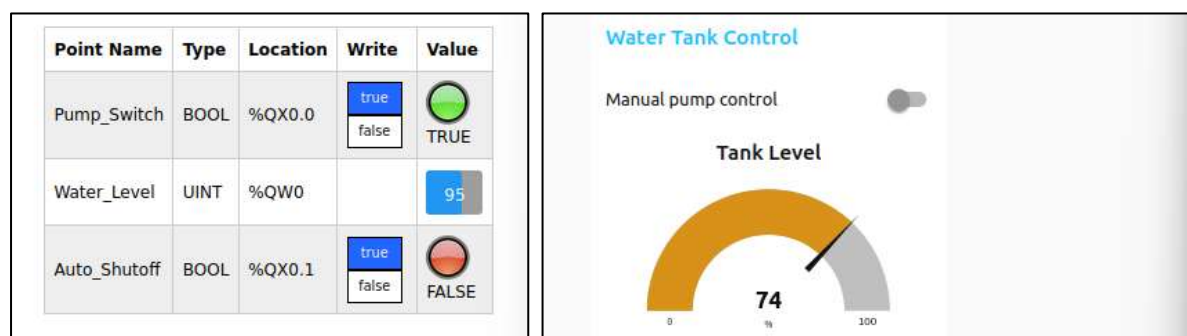
The framework successfully detected: Nmap scans, Modbus-related suspicious activity, and unauthorized communication attempts. All configured Suricata rules triggered appropriately during testing.

7.4 Performance Observations

Metric	Observation
CPU Usage	Low CPU usage during normal operation
Memory	Stable memory utilization throughout testing
Packet Monitoring	Efficient real-time packet capture and analysis
Alert Generation	Real-time alerts with minimal latency

7.5 Monitoring OpenPLC Traffic

The monitoring system successfully captured: Modbus TCP packets, PLC communication, register update traffic, and connection attempts to OpenPLC services.



7.6 Detection Results for OpenPLC

The framework detected: Nmap scans targeting OpenPLC, unauthorized Modbus communication, and suspicious port activity on port 502. Alerts were visualized in Wazuh dashboards with source and destination IP information, timestamps, and severity levels.

```
test@OpenPLC-VM:~$ sudo tail -f /var/log/suricata/fast.log
04/25/2026-07:38:01.143792  [**] [1:100002:1] MODBUS Write Single Register
(FC06) [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.200.137:
39354 -> 192.168.200.143:502
04/25/2026-07:38:01.143792  [**] [1:100008:1] Malformed Modbus Packet [**]
[Classification: (null)] [Priority: 3] {TCP} 192.168.200.137:39354 -> 192
.168.200.143:502
04/25/2026-07:38:01.144804  [**] [1:100008:1] Malformed Modbus Packet [**]
[Classification: (null)] [Priority: 3] {TCP} 192.168.200.137:39354 -> 192
.168.200.143:502
04/25/2026-09:01:45.338772  [**] [1:100008:1] Malformed Modbus Packet [**]
[Classification: (null)] [Priority: 3] {TCP} 192.168.200.137:53314 -> 192
```

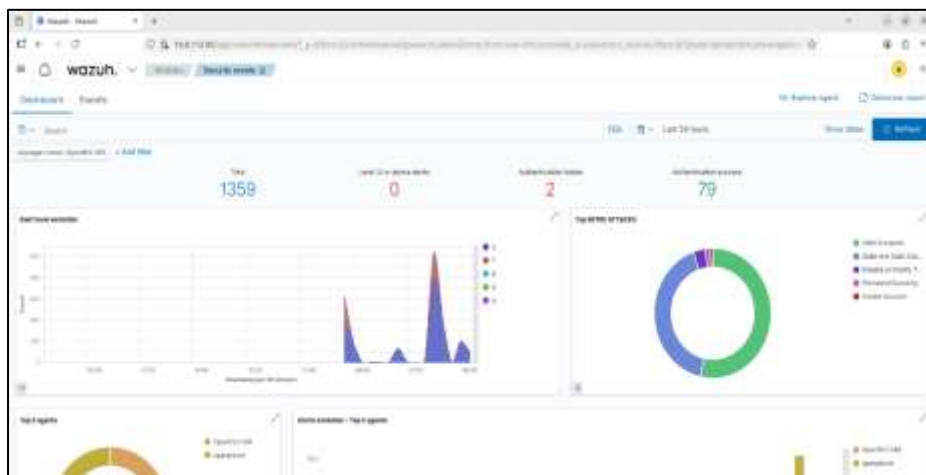
```
{
  "dest_ip": "192.168.200.143",
  "dest_port": 10606,
  "proto": "TCP",
  "flow": {
    "pkts_toserver": 1,
    "pkts_toclient": 1,
    "bytes_toserver": 60,
    "bytes_toclient": 54,
    "start": "2026-04-07T22:28:47.074733+0530",
    "end": "2026-04-07T22:28:47.074983+0530",
    "age": 0,
    "state": "closed",
    "reason": "shutdown",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  },
  "timestamp": "2026-04-07T22:29:43.268504+0530",
  "flow_id": 654839566179841,
  "n_iface": "ens33",
  "event_type": "flow",
  "src_ip": "192.168.200.137",
  "src_port": 62579,
  "dest_ip": "192.168.200.143",
  "dest_port": 46662,
  "proto": "TCP",
  "flow": {
    "pkts_toserver": 1,
    "pkts_toclient": 1,
    "bytes_toserver": 60,
    "bytes_toclient": 54,
    "start": "2026-04-07T22:28:34.925185+0530",
    "end": "2026-04-07T22:28:34.925567+0530",
    "age": 0,
    "state": "closed",
    "reason": "shutdown",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  },
  "timestamp": "2026-04-07T22:29:43.268564+0530",
  "flow_id": 1780743768463699,
  "n_iface": "ens33",
  "event_type": "flow",
  "src_ip": "192.168.200.137",
  "src_port": 62579,
  "dest_ip": "192.168.200.143",
  "dest_port": 39284,
  "proto": "TCP",
  "flow": {
    "pkts_toserver": 1,
    "pkts_toclient": 1,
    "bytes_toserver": 60,
    "bytes_toclient": 54,
    "start": "2026-04-07T22:28:40.552691+0530",
    "end": "2026-04-07T22:28:40.553469+0530",
    "age": 0,
    "state": "closed",
    "reason": "shutdown",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}
```

```
test@OpenPLC-VM:~/zeek_logs$ sudo chown $USER modbus.log
test@OpenPLC-VM:~/zeek_logs$ cat modbus.log | grep 192.168.200.143
1775584069.769355      C2x6EAUW0jnTaakf2      192.168.200.137 55690      19
2.168.200.143 502      1      1      WRITE_SINGLE_COIL      REQ      -
1775584184.071434      Cu4u0Y14aE4qThMoIf      192.168.200.137 56186      19
2.168.200.143 502      1      1      WRITE_SINGLE_COIL      REQ      -
1775584242.938047      C2YFwj1eRPbQWmmdhc      192.168.200.137 33060      19
2.168.200.143 502      1      1      WRITE_SINGLE_REGISTER  REQ      -
1775584284.197535      CMIOYZ2CTx6xlFL64      192.168.200.137 38500      19
2.168.200.143 502      1      1      WRITE_SINGLE_REGISTER  REQ      -
1775584361.859759      CoFMSK2PHubGWMfIRe      192.168.200.137 47450      19
2.168.200.143 502      1      1      WRITE_SINGLE_REGISTER  REQ      -
test@OpenPLC-VM:~/zeek_logs$
```

Chapter 8: Monitoring Interface

8.1 Wazuh Dashboard

The Wazuh dashboard provides: real-time monitoring, alert visualization, event searching, and security analytics. The dashboard serves as the central interface for all security event management.



8.2 Log Visualization

Logs are displayed using tables, alert panels, and graphical views. This improves incident analysis efficiency and enables security analysts to quickly identify patterns and anomalies.

8.3 Event Monitoring

Security analysts can monitor: active alerts, network events, suspicious activities, and PLC communication events. The interface provides real-time updates and historical event searching.

Security Alerts								
Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID	
Apr 25, 2020 @ 08:11:58.362	000	QactPLC-DM			Apparmor DENIED	5	52002	
Apr 25, 2020 @ 08:11:58.362	000	QactPLC-DM			Apparmor DENIED	5	52002	
Apr 25, 2020 @ 08:11:58.362	000	QactPLC-DM			Apparmor DENIED	5	52002	
Apr 25, 2020 @ 08:11:58.362	000	QactPLC-DM			Apparmor DENIED	5	52002	
Apr 25, 2020 @ 08:12:07.982	000	QactPLC-DM			PPM Login session closed	5	50023	
Apr 25, 2020 @ 08:12:07.982	000	QactPLC-DM	713276	Defense Evasion, Privilege Escalation, Initial Access	PPM Login session opened	5	50021	
Apr 25, 2020 @ 08:12:07.982	000	QactPLC-DM	71644-010	Defense Evasion, Defense Evasion	Successful sudo to root executed	5	54022	
Apr 25, 2020 @ 08:12:07.982	000	QactPLC-DM			Host-based anomaly detection event (network)	7	610	
Apr 25, 2020 @ 08:12:07.982	000	QactPLC-DM			Host-based anomaly detection event (network)	7	610	
Apr 25, 2020 @ 08:12:07.982	000	QactPLC-DM			Host-based anomaly detection event (network)	7	610	

8.4 Visualization of PLC Events

The dashboard displays: Modbus traffic logs, PLC-related alerts, source and destination information, and event severity levels. This provides complete visibility into industrial control system communication.

Chapter 9: Limitations and Future Scope

9.1 Simulated Environment

The project was implemented in a virtual environment and may not fully represent real industrial deployments. Physical PLC hardware, larger network topologies, and real-world industrial protocols may behave differently from the simulated environment.

9.2 Signature-based Limitations

Signature-based systems cannot detect: unknown attacks, zero-day exploits, and advanced persistent threats. This is a fundamental limitation of rule-based detection that requires supplementation with behavioral analysis or machine learning.

9.3 OpenPLC Simulation Limitations

OpenPLC provides simulation capabilities but does not fully replicate industrial-grade PLC hardware, large-scale industrial environments, or complex industrial operations involving multiple interconnected systems.

9.4 Future Improvements

Future enhancements may include:

- Machine learning-based anomaly detection
- Real industrial PLC integration (e.g., Siemens S7, Allen-Bradley)
- Advanced Modbus and DNP3 protocol anomaly detection
- Automated response systems and active countermeasures
- Integration with additional industrial protocols (OPC-UA, EtherNet/IP)
- Integration with threat intelligence feeds such as ThreatFox and AlienVault OTX
- High-availability and distributed Wazuh deployment

Chapter 10: Conclusion

10.1 Summary of Work

The project developed a CS Shield: Defensive Monitoring System for SCADA Networks using OpenPLC, Suricata, and Wazuh. The framework successfully monitored industrial traffic, detected suspicious activities, generated alerts, and visualized logs and events through the Wazuh dashboard.

The combination of OpenPLC, Suricata, and Wazuh provides a powerful, cost-effective, and open-source security monitoring solution that addresses key gaps in SCADA security: lack of visibility, absence of real-time alerting, and no centralized log management.

10.2 Learning Outcomes

The project improved understanding of: SCADA security principles, intrusion detection systems, industrial communication protocols, log monitoring and correlation, PLC traffic analysis, and virtual lab deployment techniques.

10.3 Importance of SCADA Monitoring

SCADA monitoring is essential for protecting critical infrastructure from cyber attacks and operational disruptions. The increasing convergence of IT and OT networks makes continuous security monitoring a necessity for all industrial organizations.

10.4 Importance of OpenPLC in CS Shield

OpenPLC improved the realism of the project by generating industrial communication traffic and simulating PLC operations. The integration of OpenPLC, Suricata, and Wazuh created a lightweight and effective industrial monitoring framework that can serve as a foundation for more advanced security research.

Chapter 11: Development Process

11.1 Implementation Steps

12. Environment setup — Ubuntu VMs configured
13. Tool installation — Suricata, Wazuh, OpenPLC
14. OpenPLC configuration — water tank PLC logic
15. Network configuration — virtual network adapters
16. Suricata rule configuration — custom detection rules
17. Wazuh integration — log forwarding and dashboard setup
18. Attack simulation — Nmap scans and iperf3 testing
19. Result analysis — alert review and performance measurement

11.2 Tools Configuration

Each tool was configured according to project requirements: Suricata IDS with custom rules for Nmap and Modbus detection, Wazuh for log collection and alert visualization, OpenPLC runtime for industrial simulation, and network interfaces for inter-VM communication.

11.3 Challenges and Solutions

Challenge	Solution
Log forwarding issues	Corrected permissions and Wazuh agent configurations
Interface monitoring problems	Reconfigured network adapters in promiscuous mode
False positives in rules	Modified and tuned Suricata detection rules
VM communication problems	Adjusted virtual network settings and IP routing
OpenPLC connectivity issues	Verified Modbus TCP configuration and port 502 access

References

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